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Depiction of Epidemiologic Data

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Learning Objectives

- Understand what data visualization is and why it is an essential tool in public health and epidemiologic practice
- Describe risk factors in public health and how they are measured
- Explore the Visualization Hub of the Institute for Health Metrics and Evaluation
- Describe the key components and considerations in graphing public health data

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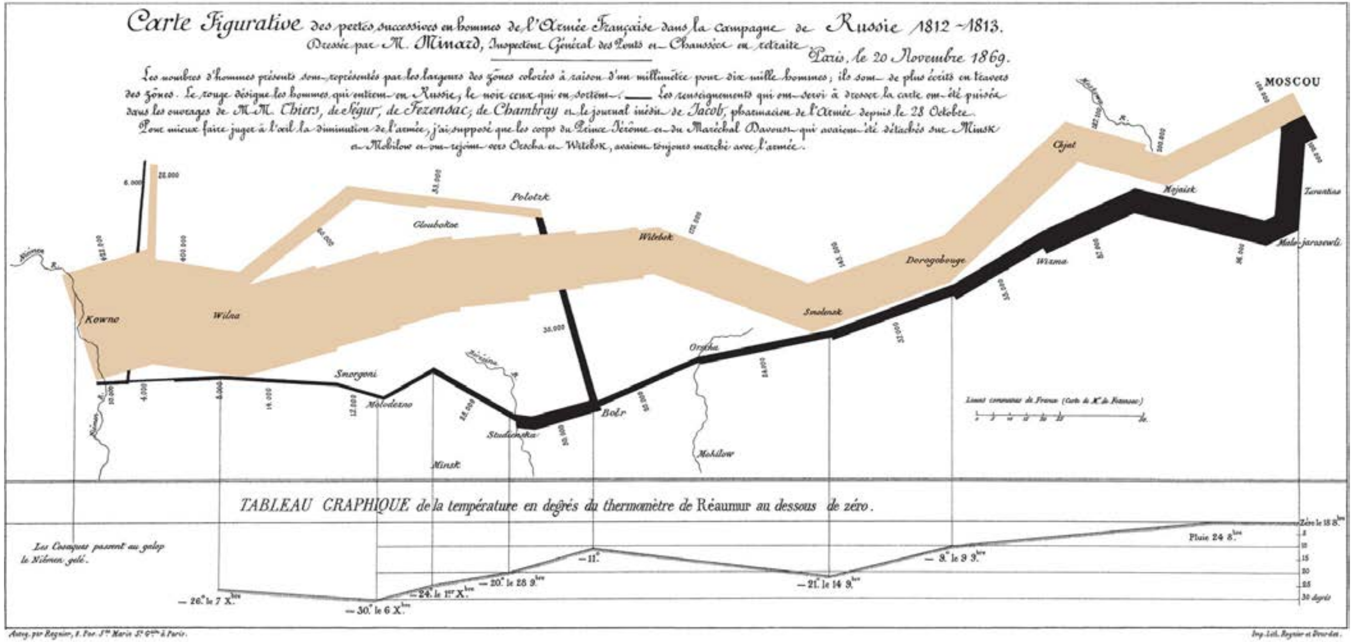
Data Visualization Definition and Basic Description

Definition: Data Visualization

- The presentation of data in pictorial or graphical format
  - Can be interactive, allowing the user to change inputs and view the resulting patterns and trends
- As computer technology has improved, data visualization has been brought to the forefront of corporate work and, more recently, of ,public health practice
- Large amounts of complex data can be better understood using visualizations as opposed to charts and tables
- “A picture is worth a thousand words”

Early Example of Data Visualization

- 1812–1813: Charles Minard (a civil engineer) created a map showing the movement of Napoleon’s army into and out of Russia
- Shows the army’s path, including number of soldiers, temperature, and time



Source: Minard, C. (1812–1813). Minard.png, via Wikimedia Commons. Public domain. Retrieved October 11, 2018.

What Data Visualization Does

- Allows individuals to comprehend information quickly and easily
- Shows patterns, relationships, and trends
- Identifies newly emerging changes, hotspots, or outliers
- Allows prediction of future events (through extrapolation, not modeling)
- Communicates a story or a message

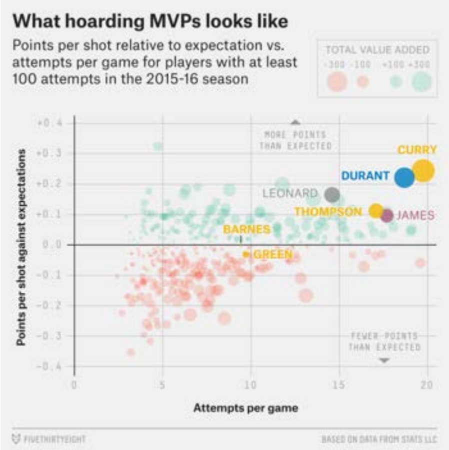
FiveThirtyEight—1

- Analyses of politics, economics, sports, and anything else of popular concern
- A licensed feature of the *New York Times* online since 2010
  - Acquired by ABC News in April 2018
- Has won numerous awards
- Look at the FiveThirtyEight website
  - Why is it so popular?

Source: Wikipedia. (2018). FiveThirtyEight. Retrieved October 11, 2018.

FiveThirtyEight—2

- One of the reasons is the creativity they use in visually depicting data related to issues that are of concern or interest to the general public



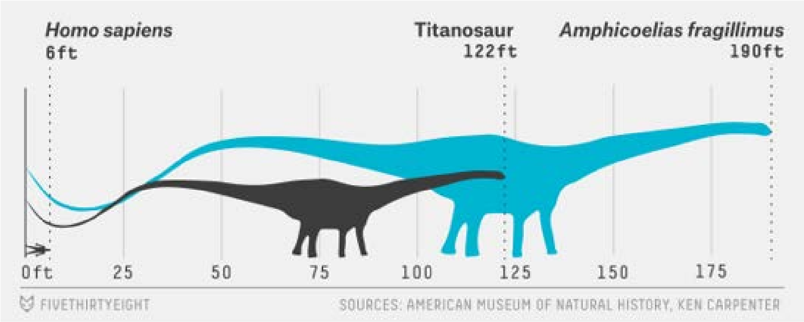


Image source: Scheinkman, A. (2016, December 30). The 52 best—And weirdest—Charts we made in 2016. *FiveThirtyEight*. Retrieved October 11, 2018.

Types of Data Visualizations

- Charts
- Graphs
- Maps
- Infographics
- Dashboards
- Storyboards

Exercise: Explore Data Visualizations—1

- The US Census Bureau compiles geographic and sociodemographic information on all US residents on a regular basis
- One of its tools is a Data Visualization Gallery
- Open one of the visualization options that is of interest to you
  - Answer the questions displayed on the following slide

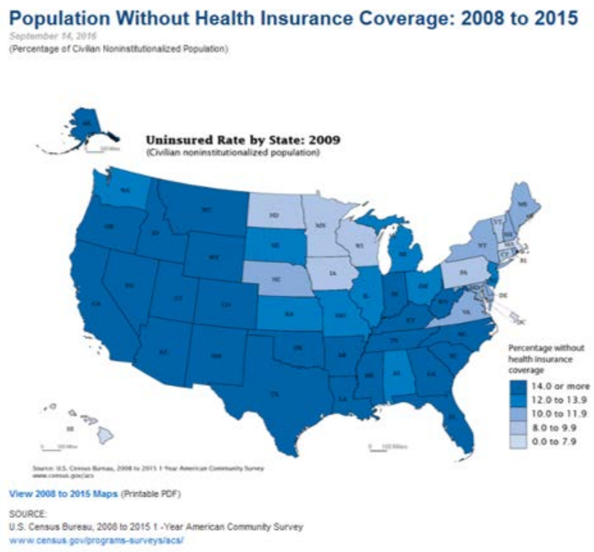
Source: US Census Bureau. (n.d.). Data: Data Visualization Gallery. Retrieved October 11, 2018.

Exercise: Explore Data Visualizations—2

- In two to three sentences, describe the data and the message that is being conveyed
- Name two positive aspects of the chosen method of visualization
- Name two areas for improvement or challenges of your chosen visualization
  - Consider: how might you have done this differently?

Exercise: Dr. Chandran's Example

- Description: this is a time-trend map showing the US population without health insurance coverage from 2008 to 2015
- Positive aspects:
  - The map is engaging as it scrolls through the relevant years in a fairly rapid fashion
  - It is easy to spot the patterns, or geographic areas where coverage is particularly good or poor
- Areas for improvement:
  - There is no clear delineation of major policy changes, like the passing of the Affordable Care Act
  - You do not see years next to each other, making it more difficult to compare



Source: US Census Bureau. (2016, September 14). Library: Infographics & Visualizations: 2016: Population without health insurance coverage: 2008 to 2015. Retrieved October 11, 2018.

Pre-steps to Creating a Data Visualization

- Understand your data

- Decide on the message or story you would like to communicate
  - This must be tailored to your target audience

- Consider logistical details
  - Should it be interactive?
  - What is the computing access and power of your target audience?
  - How current or updated can you keep your visualization?

- Learn from others—numerous blogs, books and articles, and examples are available on the Internet

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Risk Factors in Public Health: Create a Visualization

Background: Outcomes vs. Risk Factors

- Much of public health involves monitoring and studying health outcomes
  - Deaths (mortality)
  - Diseases

- An important component of public health work is the assessment of “risk factors”
  - Genetic: have a gene that predisposes a particular outcome
  - Social: crime exposure, healthy food access
  - Behavioral: tobacco use, sexual contact without protection
  - Environmental: poor water quality, air pollution
  - Health indicators: hypertension, immunizations

Risk Factor Surveillance

- Outcome data can often be collected from compilation sources such as registries, vital statistics, medical records, etc.
- Risk factor data generally needs to come from self-report by individuals
- Most countries have ongoing large surveys and surveillance efforts
- US examples:
  - Behavioral Risk Factor Surveillance System (BRFSS)
  - National Health and Nutrition Examination Survey (NHANES)

The Global Disease Burden Project—1

- Initiated in 1990 by the World Health Organization (WHO)
- Aims to provide a consistent and comparative description of the burden of diseases and injuries, and the risk factors that cause them
- Starting in 2010, the Institute for Health Metrics and Evaluation (IHME) and several academic partners began participating in the process
- Provides mortality estimates at the country, regional, and global levels (2000–2015)

The Global Disease Burden Project—2

- Method: uses vital statistics data from countries where the quality is considered useable
  - Estimates for other countries are extracted from geographically and demographically similar areas
  - Provides estimates back to the country for review and refinement

- Data are accessible from WHO and from participating organizations (IHME, etc.)

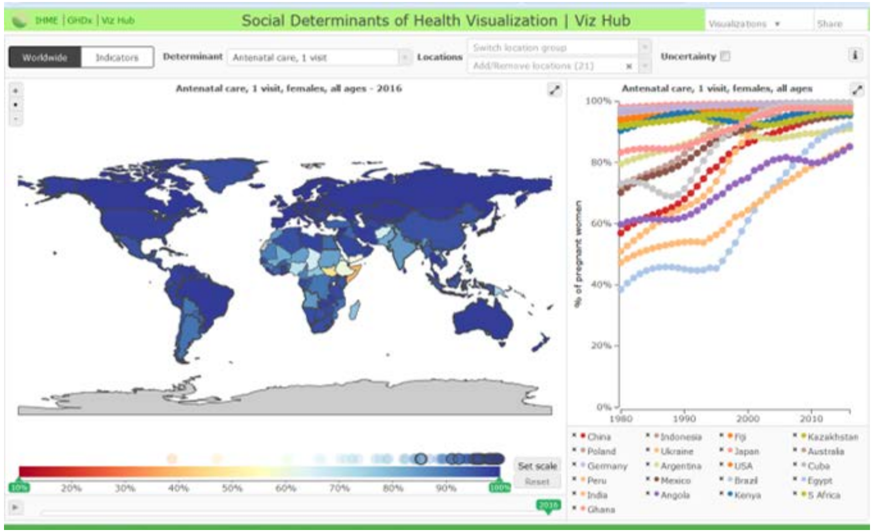
Institute for Health Metrics and Evaluation

- IHME is a population health research center
  - Part of the University of Washington
  - IHME home page link is provided on the lecture page
- IHME generates country-level risk factor estimates using known data, modeling, and statistical estimation methods
- They have a data visualization hub where this information can be explored
  - IHME Data Visualizations link is provided on the lecture page

Source: Institute for Health Metrics and Evaluation. (2018). Home page. CC BY-NC-ND 4.0. Retrieved October 11, 2018. IHME. (2018). Data Visualizations. CC BY-NC-ND 4.0. Retrieved October 11, 2018.

Explore IHME's Visualization Hub

- Go to the IHME Data Visualizations page
- Select "Social Determinants of Health Visualization"
- You can select different determinants, locations, scales, and years
- Spend some time exploring DPT3 (three doses of diphtheria, pertussis, and tetanus vaccines) immunization coverage over different locations



Exercise: Explore Global Immunization Coverage—1

- What global trends do you notice in immunization coverage over time, from 1980 onward?
- What specific pockets of lower immunization rates do you notice on the world map?
- Click on the “Uncertainty” box
  - What does that mean?

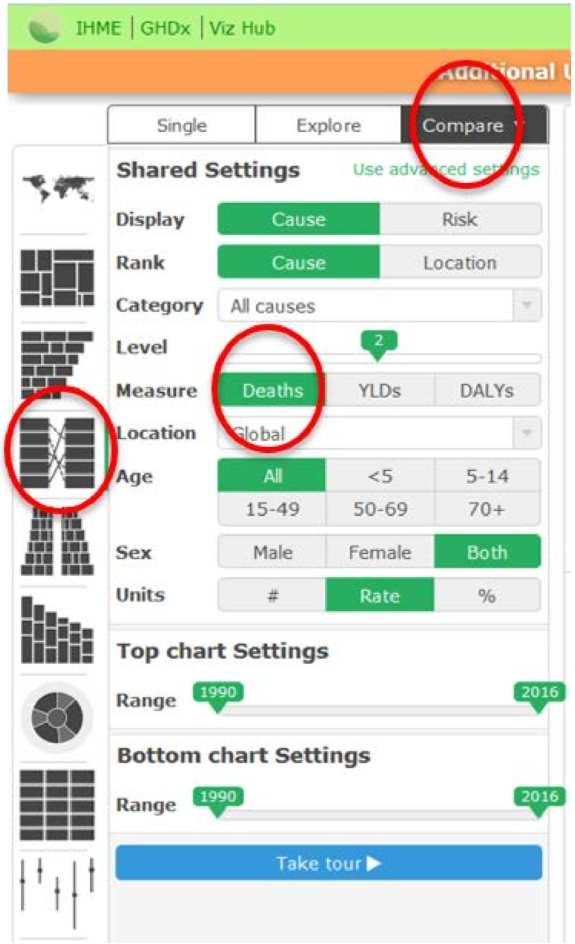
Exercise: Explore Global Immunization Coverage—2

- What global trends do you notice in immunization coverage over time, from 1980 onward?
- What specific pockets of lower immunization rates do you notice on the world map?
- Click on the “Uncertainty” box
  - Because of poor data quality, immunization coverage often has to be estimated
  - The “uncertainty” displays the mathematical uncertainty around the calculated estimates
  - You can, with some confidence (95%), say that the actual coverage lies somewhere within the shaded area surrounding each data point on the graph

Changes in Risk Factor Burden

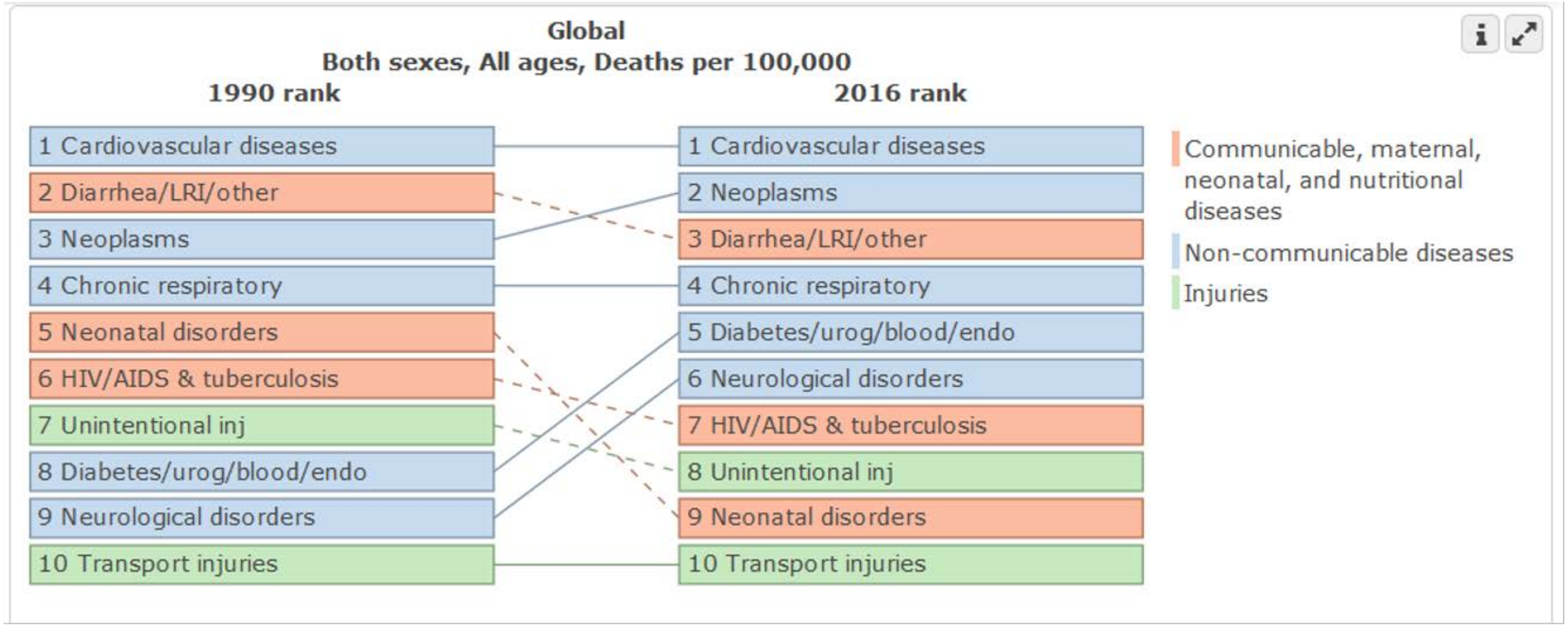
- Go back to the Data Visualizations page and select "GBD Compare"
  - IHME GBD Compare link is provided on the lecture page

- Select "Arrow diagram" on the left-most column, "Compare" on the top most row, and "Deaths" as the "Measure"



Source: IHME. (2018). GBD Compare Data Visualization. Retrieved October 11, 2018.

Risk Factor Burden Trends over Time



Source: IHME. (2018). GBD Compare Data Visualization. CC BY-NC-ND 4.0. Retrieved October 11, 2018.

Reflection: Explore Risk Factor Changes

- Hone in on risk factor burden comparisons by:
  - Location
  - Category
  - Age and sex
  - Year
- How could an arrow diagram (or another depiction of change over time) be useful in public health messaging?
- What could be some unintended consequences of seeing data displayed in this way?

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Graphs in Public Health

Graphs in Public Health: Overview

- Graphs are one of the most widely used forms of data visualization in public health and epidemiology

- Why do we use graphs?
  - Displays data in visual form
  - Can highlight patterns and differences that are harder to discern in a list of numbers
  - Effective way to communicate with a variety of audiences

- As public health professionals we have a responsibility to…
  - Convey data in a way that is accessible to our target audience
  - Convey data in a way that is NOT biased or misleading

Graphing Best Practices

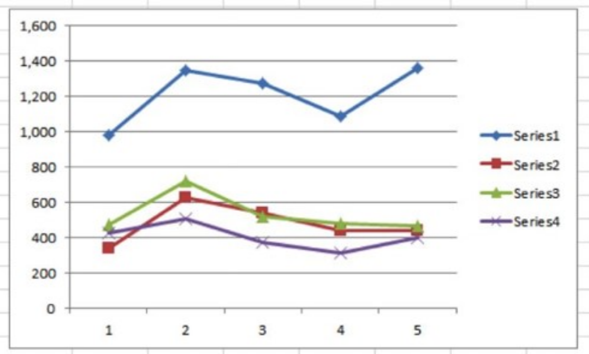
- Every line or shape should have a purpose—it should convey something
- Ensure labels and titles are all clear and self-explanatory
- Define all symbols and abbreviations
- Consider your message, and make an effort to convey this message in a straightforward and objective manner
- Decisions made in the type or style of graph can fundamentally change the story conveyed by the figure

Graphs: Important Components

- Title: clearly delineate the data being depicted
- Axis titles: let the audience know what is being shown on each axis
- Legend: describe the different components depicted on the graph
- Data source: if not indicated in the title, should be placed in a separate text box
- Essentially, the graph should be able to stand alone

Type of Graph

- Different types of graphs can highlight different points or trends in your data
- Compare the two graphs and think about the differences in the messages conveyed by each
- However, certain types of graphs may not be appropriate for certain types of data
  - Example: is it appropriate to connect the dots across the different "series" in this line graph?



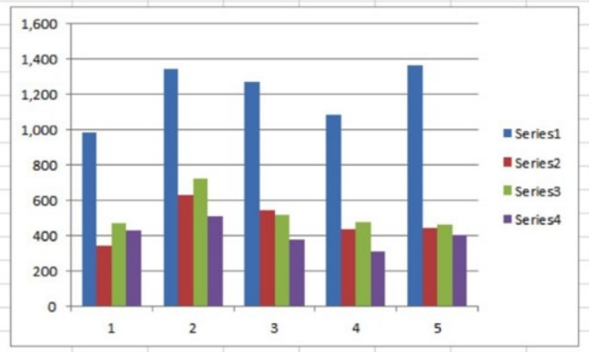
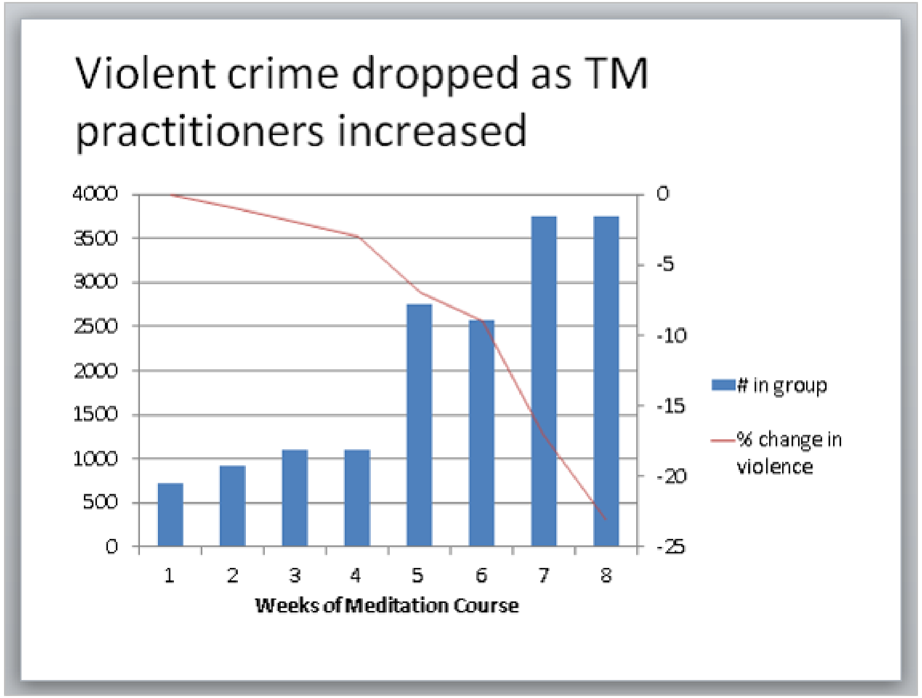


Image source: 7wData. (2015). ESCP Europe. Retrieved October 11, 2018.

Dual Axes

- When might a dual-axis graph be appropriate?
- It is a useful way to depict two very different types of data
  - Opposite trend directions
  - Different scales



Broken Axis

- Why did the graph maker choose to break the y-axis in the bottom graph?

- Clearly, the values of the first column are dwarfing the values in the subsequent categories in the top graph
  - The differences between categories 2, 3, and 4 are barely visible when the axis is unbroken

- However, consider your audience
  - Is the broken axis clear, or does it convey a misleading message?

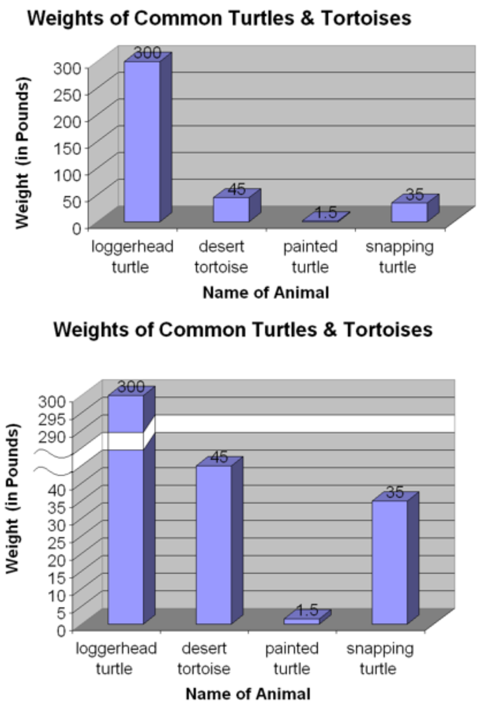
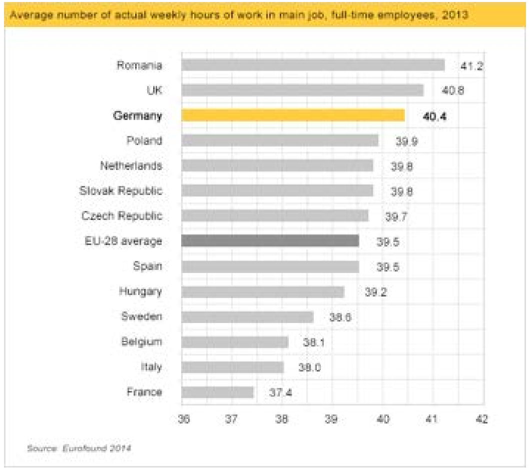


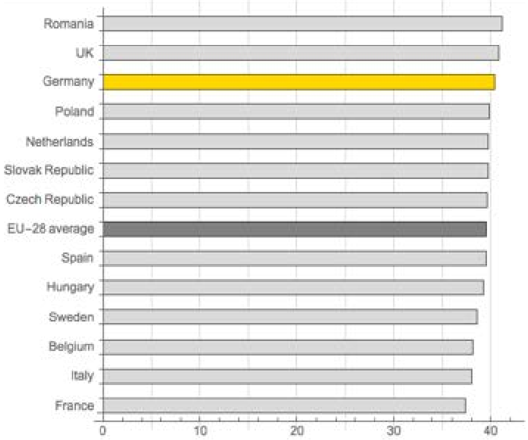
Image source: Brege, P. (n.d.). Earth Science: Middle School/Science Courses. Chapter 12, Lesson 5: Graph terminology axis, range & scale. Study.com. Retrieved October 11, 2018.

Axes: Does Starting at 0 Matter?

- In looking at the top graph, what inferences do you make about the average number of hours worked in Germany or the UK versus France or Italy?

- Now observe the bottom graph
  - Do your inferences change?



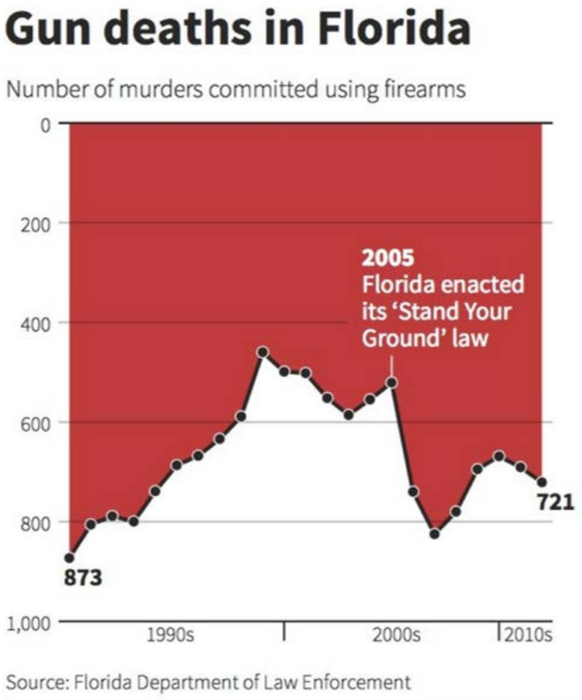


Source: Bergstrom, C., & West, J. (2017). Visualization: Misleading axes on graphs. CallingBullshit.org. Retrieved October 11, 2018.

Axes: Can They Be Inverted?

- Look at this graph
  - What inferences do you make about the impact of the highlighted legislation on firearm homicides?

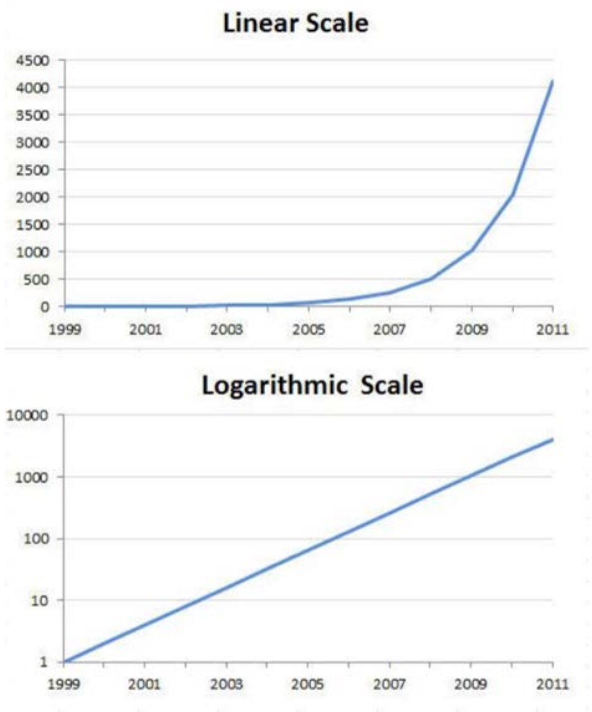
- The axis is inverted
  - Does this clearly depict an increase in the number of firearm deaths after the implemented legislation?



Source: Bergstrom, C., & West, J. (2017). Visualization: Misleading axes on graphs. CallingBullshit.org. Retrieved October 11, 2018.

Choice of Scale: Linear vs. Logarithmic

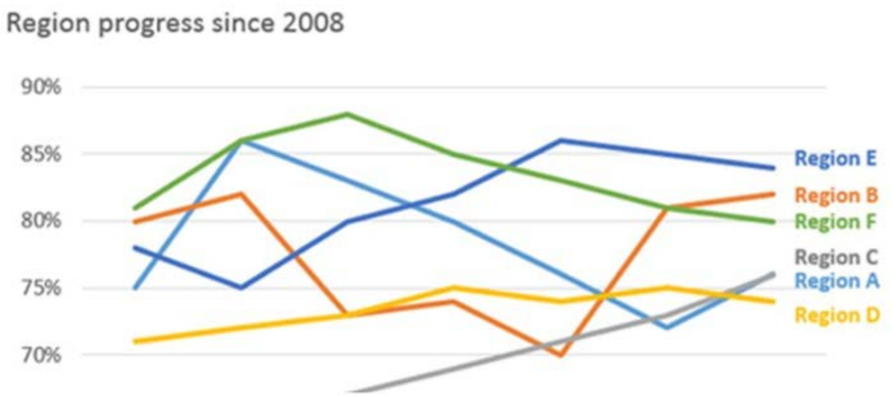
- Linear scale: depicts exact counts or rates
- Logarithmic scale: depicts counts or rates on an exponential scale
- Which graph more accurately depicts a constant rate of growth each year?



Source: Robbins, N. (2013, January 24). When should I use logarithmic scales in my charts and graphs? DataDrivenJournalism.net. Retrieved October 11, 2018.

Graphs: Color Selection

- Select colors and patterns carefully
- Are they easy to differentiate?
- Are they easy to see?
- Consider the setting in which your graph will be displayed and how the graph looks in that arena



Source: Evergreen, S. (2014). When it's OK to NOT start your axis at zero. Evergreen Data: Intentional Reporting & Data Visualization. Retrieved October 11, 2018.

Exercise: Critique a Graph—1

- Published by an economics institute in a high-income country
- Given the aspects of best practices in graph development that we have been discussing, think about some of the positive aspects of this graph and some potential areas for improvement

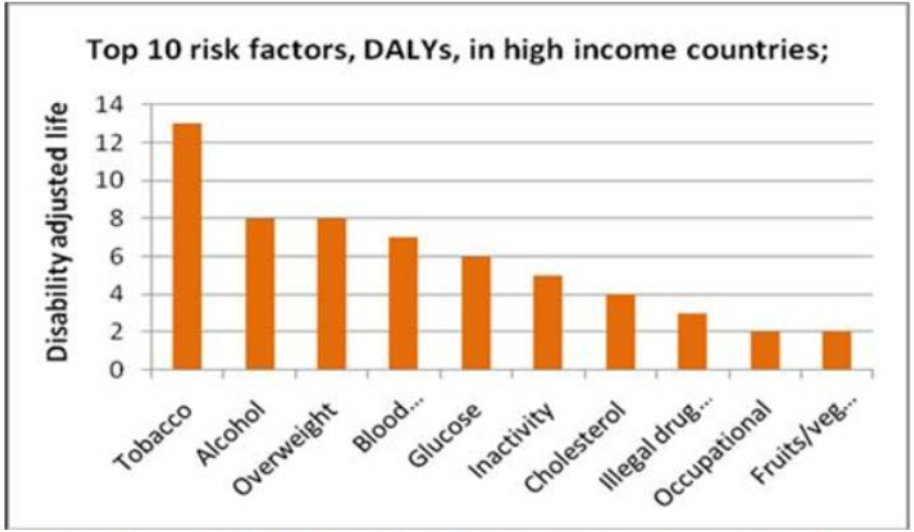
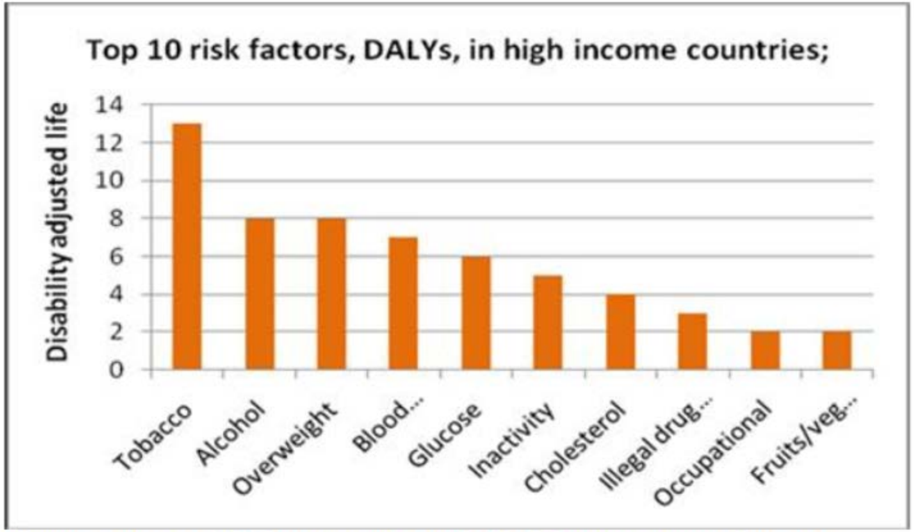


Figure 1: Top 10 risk factors, DALYs, in high-income countries, 2006
Source: World Health Organization (2008). *Global Burden of Disease*.

Source: Institute of Health Economics (IHE). (2011, September 14). Everybody's business: The cost of multi-department involvement in public health in Alberta. IHE.ca. Retrieved October 11, 2018.

Exercise: Critique a Graph—2

- Positive aspects:
  - Brings attention to risk factors, which can be actionable targets in public health practice
  - Shows in quantitative order the burden of different risk factors across high-income countries
  - Y-axis starts at zero, giving a clear depiction of the comparisons being made



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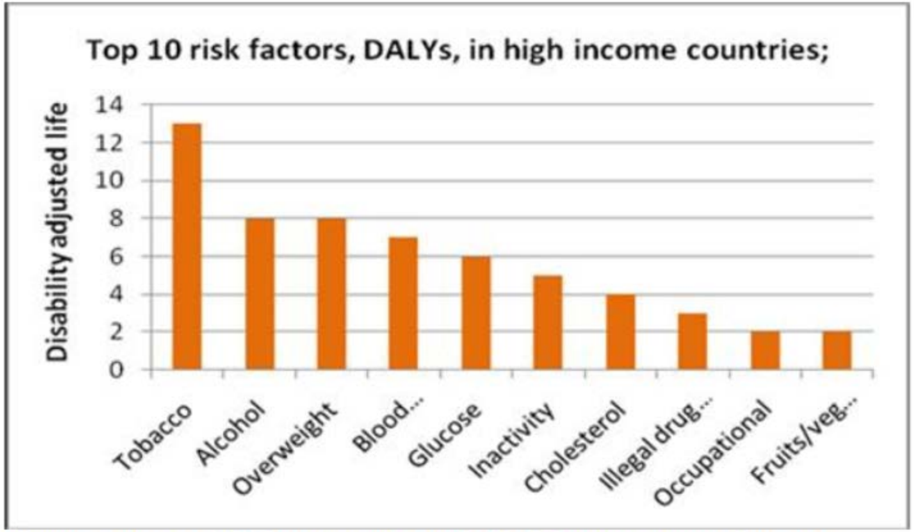
Exercise: Critique a Graph—3

- Areas for improvement:
  - The y-axis label is not clear, nor is the abbreviation used in the title: What is a DALY?
  - Several of the labels on the x-axis are cut off
  - Is the message directed to individuals (i.e., personal responsibility to change behavior) or is the message to public health officials at a community level?
    - No differentiation of access to health services, for example



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Summary

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- Graphs are one of the most widely used forms of data visualization in public health and epidemiology
- Decisions made in the type or style of graph can fundamentally change the story conveyed by the figure
- “A picture is worth a thousand words”

Thank you!